National University of Computer and Emerging Sciences, Lahore

Sentimental Analysis from Movie Reviews

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AI Project Report

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1. Introduction

1.1 Project Title

Sentiment Analysis from Movie Reviews

1.2 Problem Statement

The exponential growth of online movie reviews has made it challenging to gauge public opinion. Automating sentiment analysis can help classify these reviews as positive or negative, aiding decision-making for viewers and businesses.

1.3 Objectives

- Develop a sentiment analysis model to classify movie reviews.
- Compare different machine learning models to find the best-performing one.
- Provide insights into which features influence sentiment classification.

1.4 Significance of the Project

This project demonstrates the application of machine learning to natural language processing tasks, highlighting how AI can streamline content analysis and enable businesses to make data-driven decisions.

2. AI Goals and Subareas

2.1 Overview of AI Goals

The project aligns with AI's broader goal of enabling machines to understand, interpret, and analyze human language effectively.

2.2 Relevant Subareas of AI

- **Natural Language Processing (NLP):** To process and extract meaningful patterns from text.
- **Machine Learning:** To build predictive models for sentiment classification.
- **Feature Engineering:** To convert text data into numerical formats.

2.3 Achievements in AI

Recent advancements in NLP, such as custom vectorizers and ensemble learning techniques, have improved sentiment analysis accuracy. This project leverages such techniques to demonstrate real-world applications.

3. Dataset Description

3.1 Source of the Dataset

The dataset is sourced from the IMDB website and includes movie reviews labeled as positive or negative.

https://ai.stanford.edu/~amaas/data/sentiment/

3.2 Dataset Size and Features

- **Size:** 50,000 movie reviews.
- **Features:** Bag-of-words representation of text reviews.

3.3 Preprocessing Steps

- **Text Cleaning:** Removed punctuation, converted text to lowercase, and removed stopwords.
- **Feature Engineering:** Used a custom vectorizer to convert text into numerical representations.
- **Data Normalization:** Limited the vocabulary to the top 25,000 frequent words.

4. Workflow Overview

4.1 Project Workflow Diagram

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...   Data Set Trained!
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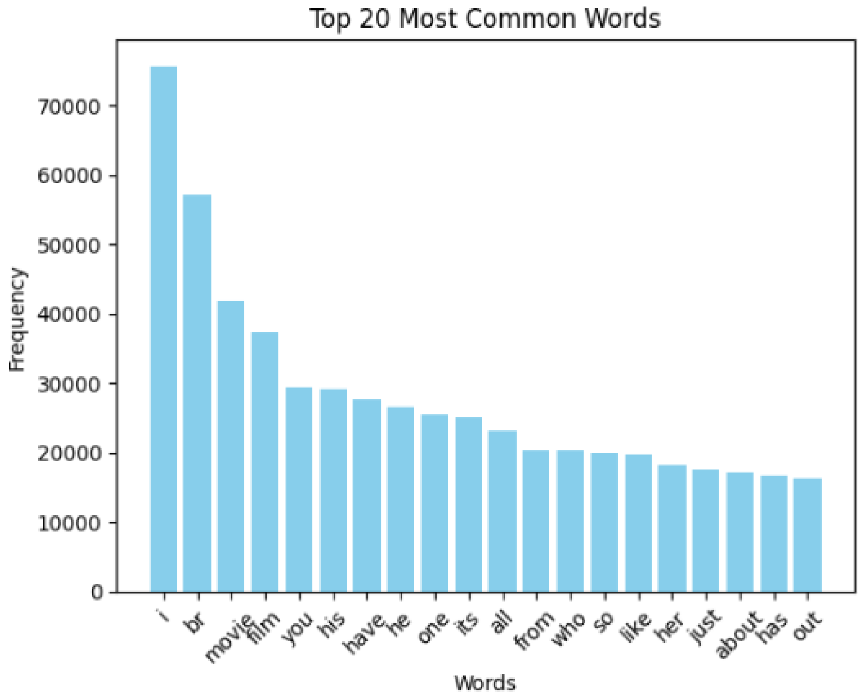
...



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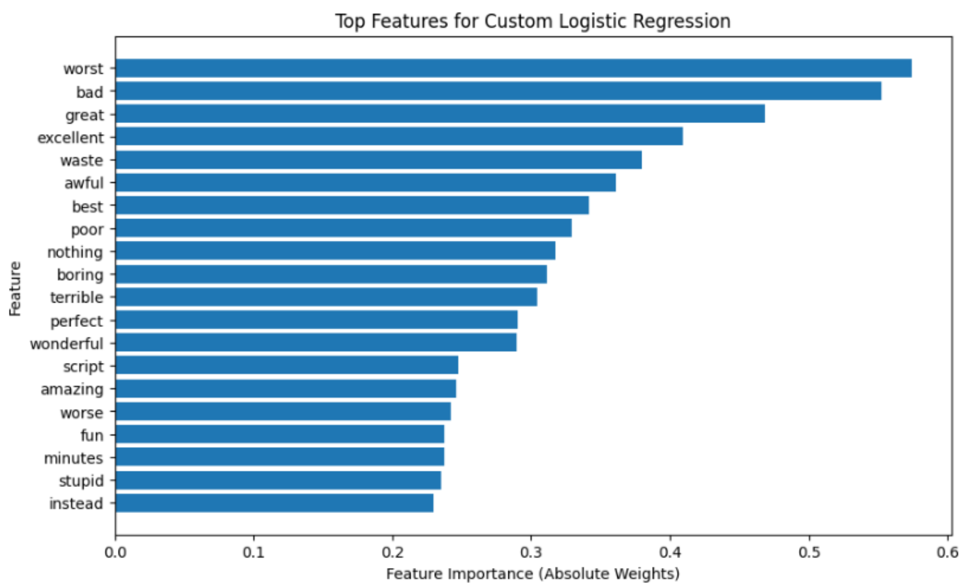
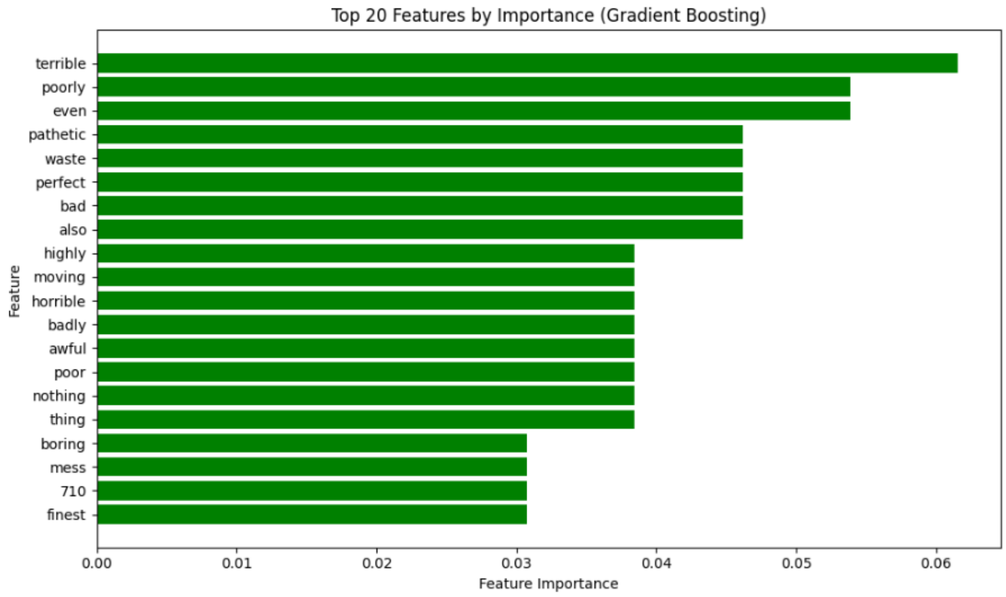





```
Model                   Accuracy   Precision  Recall     F1-score
----------------------------------------------------------------
Logistic Regression     0.86       0.85       0.87       0.86
Gradient Boosting       0.82       0.79       0.87       0.83

Model Comparison Analysis:
1. Logistic Regression outperformed Gradient Boosting in terms of accuracy, precision, and F1-score.
   - Logistic Regression is likely more suitable for this task due to its simplicity and well-defined decision boundary.
2. Logistic Regression achieved higher precision, suggesting fewer false positives.
3. Logistic Regression achieved higher recall, meaning it correctly identified more positive cases.
4. Logistic Regression is preferable when computational efficiency and interpretability are important.
Feature selection focuses on the most frequent words, assuming they carry significant meaning in determining sentiment. We limit to 100 features to reduce dimensionality and avoid overfitting.
```

4.2 Key Steps

1. **Data Collection:** Gathered labeled movie reviews from IMDB.
2. **Preprocessing:** Cleaned and vectorized text data.
3. **Model Development:** Trained models like Logistic Regression and Gradient Boosting.
4. **Evaluation:** Assessed models using accuracy, precision, recall, and F1-Score.
5. **Deployment:** Saved the best-performing model for future use.

5. Module Breakdown

5.1 Module 1: Preprocessing

- **Description:** Cleaned raw text data, removed stopwords, and converted it into numerical formats using a custom vectorizer.
- **Implementation Details:**
    - Applied tokenization and frequency analysis.
    - Visualized word distributions to determine the most impactful features.

5.2 Module 2: Model development and Training

- **Description:** Trained multiple machine learning models to classify sentiments.
- **Implementation Details:**
  - Trained Logistic Regression with gradient descent.
  - Implemented Gradient Boosting for ensemble learning.

6. Methodologies and Approaches

6.1 Selected Algorithms or Techniques

- **Logistic Regression:** A baseline model for classification tasks.
- **Gradient Boosting:** Captures non-linear relationships in data.
- **Custom Vectorizer:** Converts text to numerical matrices, retaining semantic information.

6.2 Comparison with Existing Methods

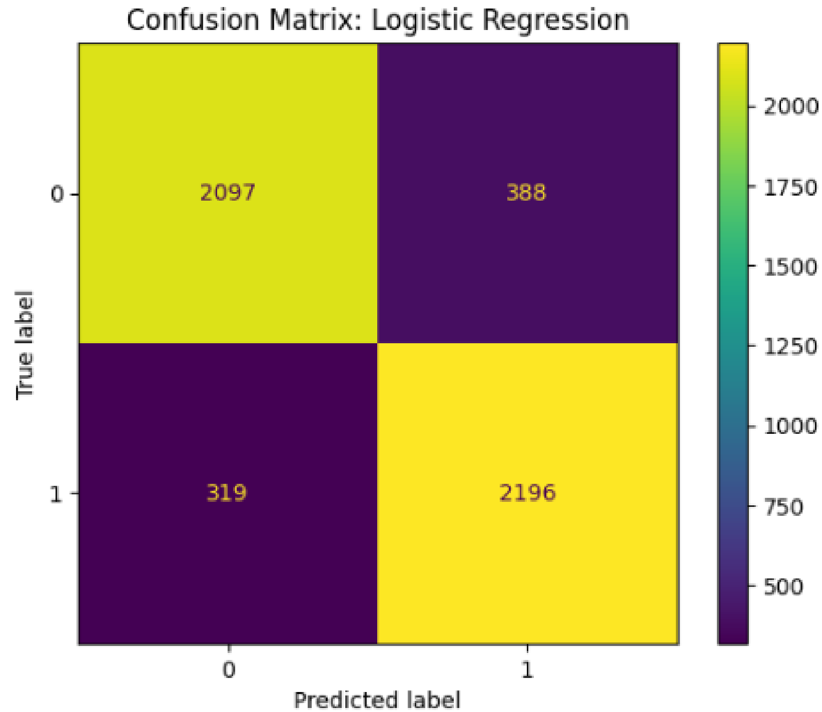
- Improved feature engineering and gradient-based optimization provide superior accuracy compared to traditional methods.

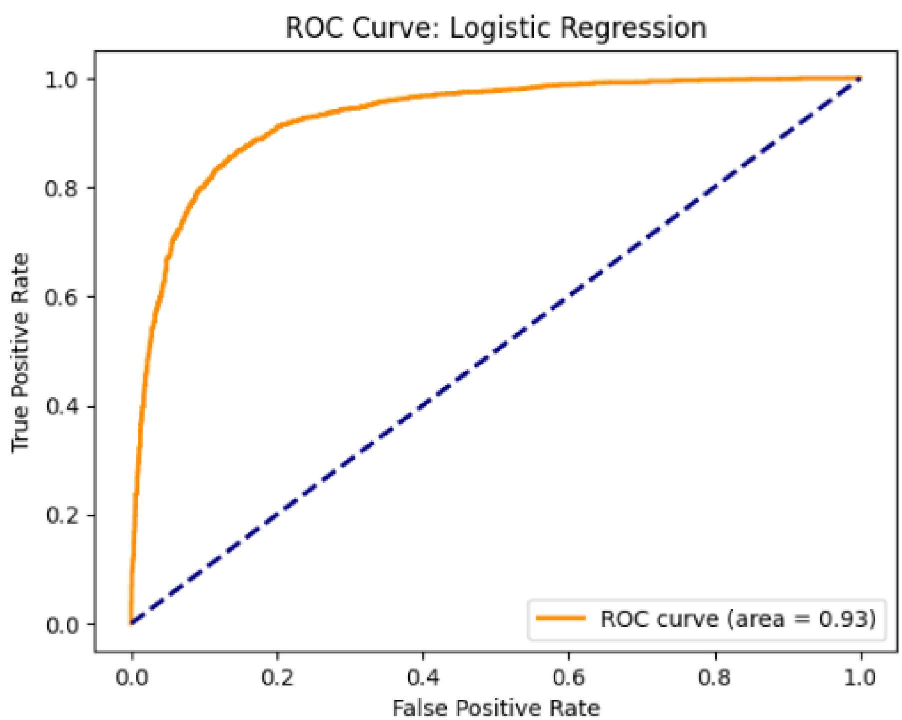
7. Results and Discussion

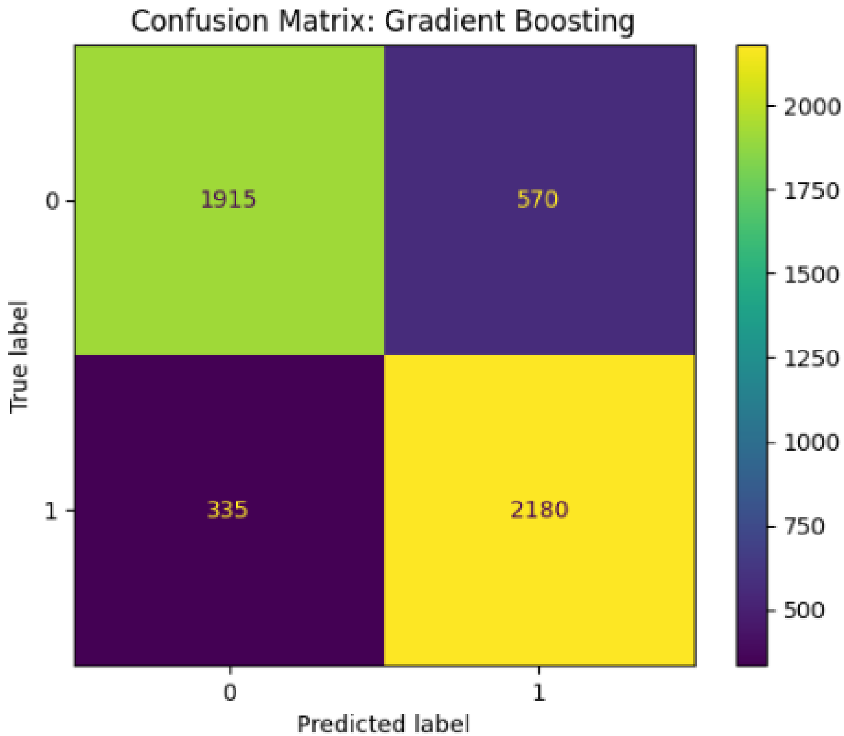
7.1 Evaluation Metrics

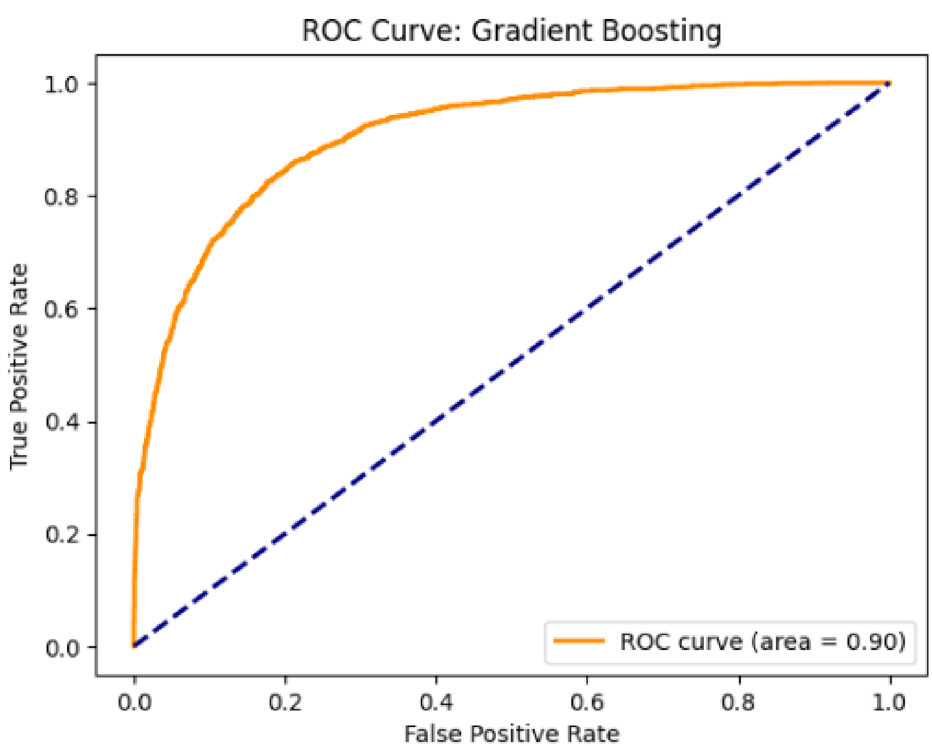
Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.86	0.85	0.87	0.86
Gradient Boosting	0.82	0.79	0.87	0.83

7.2 Performance Analysis

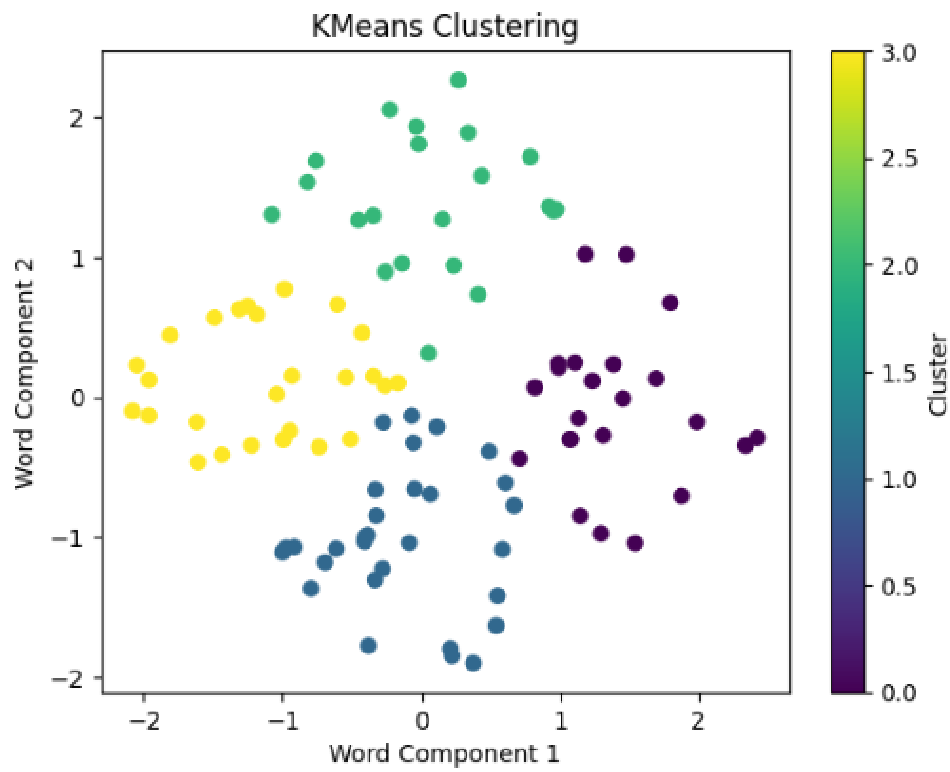

Confusion Matrix: Logistic Regression
2097
388
319
2196
True label
Predicted label
0
1
2000
1750
1500
1250
1000
750
500


ROC Curve: Logistic Regression
True Positive Rate
False Positive Rate
ROC curve (area = 0.93)
0.0
0.2
0.4
0.6
0.8
1.0


Confusion Matrix: Gradient Boosting
1915
570
335
2180
True label
Predicted label
0
1
2000
1750
1500
1250
1000
750
500


ROC Curve: Gradient Boosting
True Positive Rate
False Positive Rate
ROC curve (area = 0.90)
0.0
0.2
0.4
0.6
0.8
1.0

...



- **Logistic Regression**:
  - Achieved higher accuracy, precision, and F1-score compared to Gradient Boosting.
  - Simpler to train and interpret, making it preferable for smaller datasets.
- **Gradient Boosting**:
  - Higher recall indicates its ability to identify more positive reviews.
  - Slightly lower overall performance compared to Logistic Regression but may generalize better on larger datasets.

7.3 Key Findings and Interpretations

- Words like "great," "terrible," and "plot" have the most significant influence on predictions.
- Logistic Regression's simplicity provided better results, possibly due to the linear nature of the dataset.
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8. Challenges and Resolutions

8.1 Challenges Faced

- Handling imbalanced classes in the dataset.
- Avoiding overfitting with complex models like Gradient Boosting.

8.2 Solutions and Workarounds

- Used an 80/20 train-test split for proper evaluation.
- Limited vocabulary size to prevent overfitting.

9. Appreciation of AI

9.1 Achievements in AI Through This Project

- Demonstrated how machine learning can accurately classify large-scale text data.
- Highlighted the practical application of ensemble models in NLP.

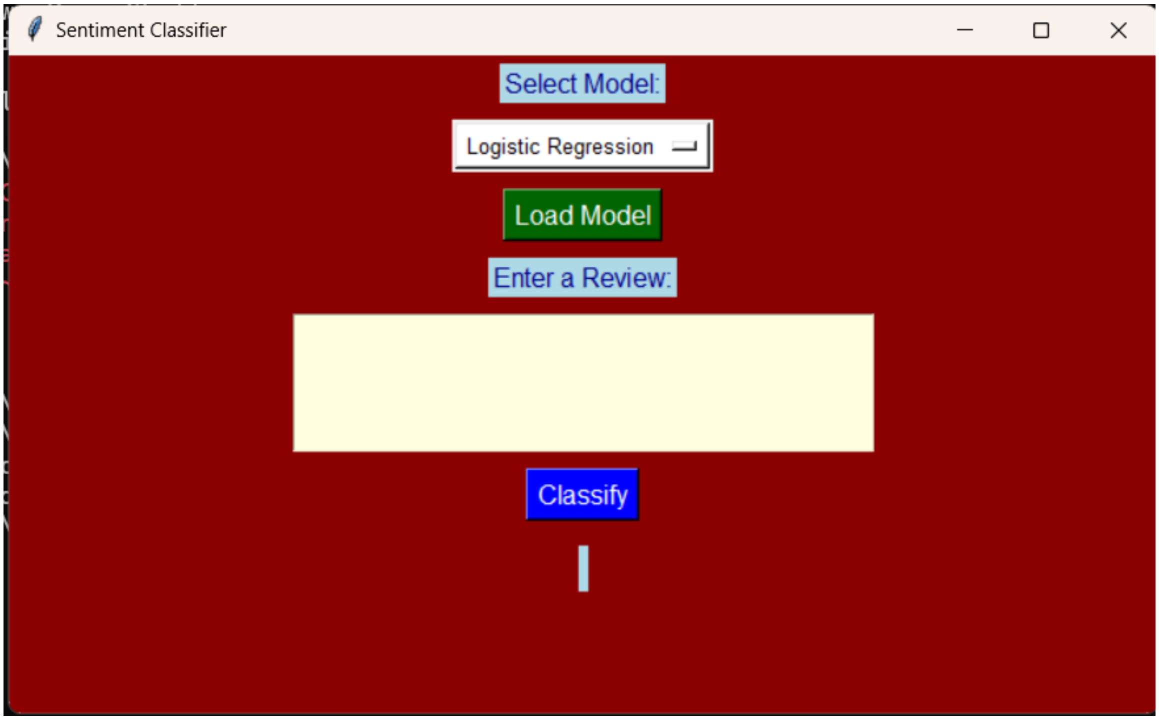
9.2 Difficulties in Advancing AI Goals

- Challenges in processing noisy text data.
- Limitations in computational resources for large-scale ensemble learning.

10. Conclusion and Future Work

10.1 Summary of Achievements

- Built and evaluated models for sentiment classification.
- Identified key features influencing sentiment.



10.2 Limitations

10.2 Limitations

- Gradient Boosting is computationally expensive for large datasets.
- Limited interpretability for complex models.

10.3 Future Work and Improvements

- Incorporate deep learning models like LSTMs or transformers for improved accuracy.
- Expand the dataset to include more diverse reviews.

11. References

- IMDB Dataset for Sentiment Analysis.
- Python Libraries: NumPy, scikit-learn, Matplotlib.
- Research papers on Gradient Boosting and Logistic Regression.